

Unitary equivalence without an initial unitality assumption

MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion_unitaryEquivalent_nonUnital

theorem

Shows why allowing nonunital input representations does not enlarge the irreducible representation class of the shell-generated algebra.

Statement

Let $T \subseteq B(\mathcal{H})$ be the fixed shell-generated algebra below. If $\rho : T \rightarrow B(K)$ is a nonzero irreducible complex-linear star representation, then a surjective complex-linear isometry $U : \mathcal{H} \rightarrow K$ satisfies $U(\rho_0(t)x) = \rho(t)(Ux)$ for every $t \in T$ and $x \in \mathcal{H}$. Preservation of the unit is a consequence, not an additional hypothesis on ρ .

Assumptions

Let A be the completed CAR algebra with matrix stages $M_{2^n}(\mathbb{C})$ embedded by $a \mapsto a \otimes I_2$. Write f_n for the image of the first diagonal matrix unit, $p_n = f_n - f_{n+1}$, and ω for the root state characterized by $\omega(\iota_n(a)) = a_{00}$ on each stage, where ι_n is its canonical embedding.

Index chosen pure-state Gelfand-Naimark-Segal (GNS) representations by their unitary-equivalence classes $j \in I$, with root index o and representative states φ_j .

A fixed shell family consists of complex-linear unital star automorphisms α_j and elements $w_{j,n} \in A$ satisfying $\varphi_j(\alpha_j(a)) = \omega(a)$, $w_{j,n}^* w_{j,n} = \alpha_j(p_n)$ and $w_{j,n} w_{j,n}^* = p_n$.

At the root, α_o is the identity and $w_{o,n} = p_n$.

Let $\mathcal{H} = \bigoplus_{j \in I} H_j$ be the Hilbert direct sum of these GNS spaces, $\pi = \bigoplus_j \pi_j$ their representation of A , and $b_j = e_j \xi_j \in \mathcal{H}$ its embedded unit cyclic vector, where $\xi_j \in H_j$ is the selected unit cyclic vector and $e_j : H_j \rightarrow \mathcal{H}$ is coordinate inclusion.

Choose once a family of unitary operators $L_j \in B(\mathcal{H})$ from the construction of the shell unitaries: $L_j b_j = b_o$, $L_j \pi(\alpha_j(p_n)) = \pi(w_{j,n})$, and $L_o = I_{\mathcal{H}}$. Here $B(\mathcal{H})$ denotes the bounded complex-linear operators on $\mathcal{H}$.

Put $T = C^*(\pi(A), \{L_j : j \in I\}) \subseteq B(\mathcal{H})$, the norm-closed unital star algebra generated by these operators. Let $\iota : A \rightarrow T$ be $\iota(a) = \pi(a)$ with its codomain restricted to T , and let $\rho_0 : T \rightarrow B(\mathcal{H})$ be inclusion.

The space K is any complete complex Hilbert space in an independent universe. Irreducibility here explicitly includes $\rho \neq 0$ and the assertion that its closed reducing subspaces are only 0 and K . Nontriviality of K alone is not a replacement for $\rho \neq 0$.

Conclusion

The intertwining equation uses the original map ρ . All its represented operators are retained when one regards it as a unital representation. Thus the displayed inclusion represents every such irreducible class up to unitary equivalence.

The result imposes no separability restriction and does not claim that all representations are irreducible. The shell family is fixed before K is chosen. The zero representation is excluded by the exact irreducibility predicate.

Proof route

Put $P = \rho(1)$. It is an idempotent, and its range is closed and reducing for ρ . Nonzeroness makes this range nonzero, so irreducibility makes it all of K ; an idempotent is the identity on its range, giving $P = I_K$. The unital capture theorem applies to the same operators, since the unit law has just been established. Its unitary intertwiner therefore satisfies the equation for the original map.

▼ Proof steps

1. For every t , $\rho(1)\rho(t) = \rho(t) = \rho(t)\rho(1)$. Star preservation also controls adjoints, so the range of $\rho(1)$ is reducing.
2. If $\rho(1) = 0$, then every $\rho(t)$ is zero. Irreducibility therefore forces the closed range of this nonzero idempotent to be K , whence $\rho(1) = I_K$.
3. Use the theorem identifying every unital irreducible representation with the fixed inclusion. Rebundling ρ as unital changes no operator, so its resulting unitary equivalence is exactly the claimed one.

Main citations

- Exact statement and proof
- A fixed family of compatible shell data
- Construction of the shell unitaries
- Every irreducible representation is equivalent to the faithful inclusion
- Capture including nonunital irreducible representations
- Irreducibility forces preservation of the unit
- The root state bundled with its purity proof

Lean source signature (exact)

```
theorem shellFamilyInclusion_unitaryEquivalent_nonUnital
  (family : RepresentativeShellFamily)
  {K : Type v} [NormedAddCommGroup K] [InnerProductSpace ℂ K]
  [CompleteSpace K]
  (rho : NonUnitalRepresentation
    (A := ShellFamilyTarget family) (H := K))
  (hrho : rho.IsIrreducible) :
```

```

    ∃ U :
      MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
    ≈i[ $\mathbb{C}$ ] K,
    ∀ (a : ShellFamilyTarget family)
      (x : MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
        completedRootPureState),
      U (shellFamilyInclusion family a x) = rho a (U x)

```

In the source	Mathematical meaning
family : RepresentativeShellFamily; ShellFamilyTarget family	The supplied normalized CAR shell family fixes once the links L_j and target $T = C^*(\pi(A), \{L_j\})$. The algebra T itself is unital.
K; [InnerProductSpace $\mathbb{C}$ K]; [CompleteSpace K]	Any complete complex Hilbert space K in the independently quantified universe, without a separability or dimension restriction.
rho : NonUnitalRepresentation (A := ShellFamilyTarget family) (H := K)	The complex-linear star representation $\rho : T \rightarrow B(K)$ is not initially required to preserve the unit.
hrho : rho.IsIrreducible	The original ρ is nonzero and has only 0 and K as closed reducing subspaces. Merely assuming $K \neq 0$ would not exclude the zero representation.
∃ U : SelectedAtomicHilbert completedRootPureState ≈ _i [$\mathbb{C}$] K	One surjective complex-linear isometry $U : \mathcal{H} \rightarrow K$ is produced, where $\mathcal{H} = \bigoplus_j H_j$ is the fixed selected atomic Hilbert sum.
∀ (a : ShellFamilyTarget family) (x : SelectedAtomicHilbert ...), U (shellFamilyInclusion family a x) = rho a (U x)	For every $t \in T$ (source a) and $x \in \mathcal{H}$, the same U obeys $U(\rho_0(t)x) = \rho(t)(Ux)$, with ρ_0 literal inclusion. The equation retains the original map ρ ; its unit law $\rho(1) = I_K$ is derived, not an extra input.

Further source notes: Here `completedRootPureState` is the root state ω , bundled with its purity proof, and `SelectedAtomicHilbert completedRootPureState` is $\mathcal{H}$. `NonUnitalRepresentation` means a complex-linear star representation without a required unit law. The existential quantifier supplies the unitary U in the Statement; the proof first derives $\rho(1) = I_K$.

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▼ Exact content identity

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